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Batch 15

Algorithm

1. Start.
2. Read the size `size` of the array.
3. If `size <= 0`, display an error and terminate.
4. Declare the variable-length integer array `arr[size]`.
5. Read the elements of `arr[]` in ascending order.
6. Read `target`.
7. Pass the address of `arr[]`, `size`, and `target` to `ceil_index()`.
8. In `ceil_index()`:
   - Traverse the sorted array from index `0`.
   - Find the first element for which `arr[i] >= target`.
   - Return its index immediately.
   - If no such element exists, return `-1`.
9. In `main()`:
   - If the returned index is `-1`, display `-1`.
   - Otherwise, display the ceil element and its index.
10. Stop.

Test Case Table

Test	Array	Target	Expected Output
1	5 7 7 8 8 10	6	Ceil of 6: 7 at index 1
2	5 7 7 8 9 10	8	Ceil of 8: 8 at index 3
3	5 7 7 8 8 10	11	-1
4	5 7 7 8 8 10	7	Ceil of 7: 7 at index 1
5	5 7 7 8 8 10	4	Ceil of 4: 5 at index 0
6	5 7 7 8 8 10	10	Ceil of 10: 10 at index 5
7	5 7 7 8 8 10	9	Ceil of 9: 10 at index 5

C program

```
#include <stdio.h>

/* Function to find the index of ceil element */
int ceil_index(int *arr, int size, int target)
{
    int i;

    for (i = 0; i < size; i++)
    {
        if (*(arr + i) >= target)
        {
            return i;
        }
    }

    return -1;
}

int main()
{
    int size, i, target, index;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid array size.\n");
        return 0;
    }

    /* Variable-length array */
    int arr[size];

    printf("Enter %d elements in ascending order:\n", size);

    for (i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter target: ");
```

```
    scanf("%d", &target);

    /* Passing the address of the array */
    index = ceil_index(arr, size, target);

    if (index == -1)
    {
        printf("-1\n");
    }
    else
    {
        printf("Ceil of %d: %d at index %d\n",
               target, *(arr + index), index);
    }

    return 0;
}
```

Execution

